Beyond the Reranker: Do RAG Retrieval Enhancements Help Once a Strong Reranker Is Present?

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Abstract

Retrieval-augmented generation (RAG) is routinely extended with methods meant to improve retrieval: query expansion, hierarchical and cross-document summarization, graph-based expansion, per-query routing, rank fusion, and corrective re-retrieval. The benefits reported for these methods come almost exclusively from homogeneous corpora, predominantly Wikipedia prose. Whether they hold on the mixed-format collections common in practice, where code, markdown, tables, scientific PDFs, and prose are interleaved within one corpus, has not been measured. To study this directly, we build **HetDocQA**, a heterogeneous benchmark with *chunker-agnostic* span-overlap relevance labels and collection-disjoint splits, and pair it with MuSiQue and QASPER as homogeneous controls. We evaluate eight methods on a shared backbone, with bootstrap confidence intervals and multiple-comparison correction. A strong cross-encoder reranker accounts for most of the pipeline's quality; beyond it, only two methods yield reliable gains: query expansion and SSCC. SSCC, a per-source calibrated corrector introduced here, sets a separate acceptance threshold for each score source and helps only on heterogeneous data. The remaining reranking and pool-expansion methods in common use, among them hierarchical summarization, graph expansion, routing, and rank fusion, give no reliable gain once that reranker is present.

1 Introduction

Retrieval-augmented generation (RAG) grounds a language model's answers in passages drawn from an external corpus [1, 2, 3]. The retrieval stage that supplies those passages has become the target of a large and growing toolkit: methods that rewrite or expand the query [4, 5, 6], summarize the corpus into hierarchies [7] or link it into graphs [8, 9], grade the retrieved evidence and retrieve again when it is weak [10, 11, 12], and route or fuse several retrieval strategies [13, 14]. Each is reported to raise end-to-end accuracy, and all are now in wide use.

This toolkit, however, has been validated on a narrow slice of the retrieval landscape. Almost all of the supporting evidence comes from homogeneous prose: open-domain and multi-hop question answering over Wikipedia [15, 16, 17, 18], and, less often, over individual scientific documents [19, 20]. The narrowness persists as the toolkit grows. Graph, routing, reranking, query-rewriting, and hierarchical methods introduced in 2025 and 2026 still report results on the same Wikipedia-derived collections, among them Natural Questions, TriviaQA, PopQA, HotpotQA, 2WikiMultiHopQA, and MuSiQue [21, 22, 23, 24, 25, 26, 27, 28, 29]. Recent surveys note the same concentration [30, 31].

Practical retrieval targets are seldom so uniform. A working collection mixes source code, markdown documentation, spreadsheets, and PDFs containing tables and equations, and the evidence for a single question is often distributed across several such documents [32]. Two questions then remain open: whether the toolkit transfers to such heterogeneous collections at all, and, if so, at which stage of the pipeline the gains arise.

This paper addresses both questions. HetDocQA is a question-answering benchmark whose collections interleave code, markdown, prose, tables, and scientific PDFs. Its relevance labels are character spans in the source documents, matched to a system's own chunks at evaluation time, so a retrieval score does not depend on how a system segments the corpus; its splits are disjoint by collection, so a threshold tuned on development data cannot exploit structure that recurs in the test set. Eight representative methods are combined in one pipeline, isolated one at a time, and run on HetDocQA together with MuSiQue and QASPER as homogeneous controls, with the embedder, reranker, and generator held fixed across all conditions. The size of the comparison grid makes correction for multiple comparisons necessary, and an effect is treated as reliable only if it remains significant after that correction. Many of these methods are specified only partially in prior work, so Section 4 presents each in full, with the hypothesis under which it should help on heterogeneous data.

Across the three benchmarks, two methods yield improvements that are statistically significant after correction (Figure 1). Query expansion helps when a question and its evidence share little surface vocabulary, a situation heterogeneous collections make common. SSCC, the per-source calibrated corrector introduced here, sets a separate acceptance threshold for each score source, since candidates from different sources and modalities arrive on incomparable scales; it improves results on HetDocQA and has no effect on the homogeneous controls. The remaining methods, which rerank or expand the candidate pool (hierarchical summarization, the

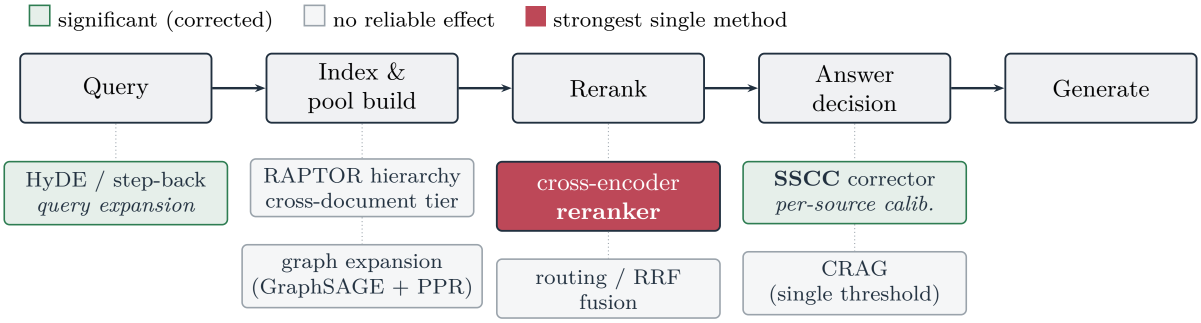


Figure 1: Methods by pipeline stage. Each method evaluated in this study is drawn under the stage of the shared pipeline it acts on, and shaded by whether its effect on HetDocQA is significant after Holm–Bonferroni correction (Section 6). The two methods with a reliable effect act on the query (HyDE) and at the answer decision (SSCC). Those that rerank or expand the candidate pool (hierarchical summarization, the cross-document tier, graph expansion, routing, rank fusion, and corrective re-retrieval) show no reliable effect. The cross-encoder reranker, shaded separately, has the largest effect of any single method.

cross-document tier, graph expansion, routing, and rank fusion), show no reliable improvement once a strong cross-encoder reranker is in place.

Contributions.

1. **HetDocQA**, a heterogeneous multi-format question-answering benchmark spanning code, markdown, prose, tabular, and PDF sources, with chunker-agnostic span-overlap relevance labels and collection-disjoint splits, released with a datasheet, deterministic build scripts, and an evaluation script (Section 3).
2. Evidence on which widely-used retrieval methods help on heterogeneous documents, from a shared-backbone comparison across HetDocQA and two homogeneous benchmarks. The cross-encoder reranker is the single largest effect; beyond it, among the established enhancements only query expansion yields a reliable gain, while the reranking and pool-expansion methods in common use yield none once that reranker is present (Sections 6 and 7).
3. **SSCC**, a per-source calibrated corrector for the answer-decision stage; it yields a reliable gain on HetDocQA and no effect on the homogeneous controls (Section 6).

All eight methods are implemented on the shared backbone and released with the benchmark, the per-query results, and the build and evaluation scripts that reproduce the full study (Section 4).

2 Related Work

Methods for the retrieval stage. The methods examined here come from four lines of work on improving retrieval for RAG, organized in Section 4 by the stage they act on. Query-side methods reformulate the question, generating a hypothetical answer to retrieve against [4, 6] or abstracting it to a more general form [5]. Index-side methods add structure to the corpus: RAPTOR [7] builds a tree of recursive summaries, and GraphRAG [8] and HippoRAG [9] build and traverse entity graphs. Decision-side methods inspect the retrieved evidence and act on it, grading and re-retrieving when it is weak [10], learning to critique [11], or interleaving retrieval with generation [12]. A fourth line selects or combines retrieval strategies per query, through routing [13] or rank fusion [14]. Each is reported to improve accuracy on the benchmark it was introduced with, and work in 2025 and 2026 continues to extend these families along the same lines [21, 23, 24, 25, 26, 27, 28]. The evaluations behind these claims share a homogeneous-prose setting, which is the assumption this paper tests.

Re-examining reported gains. A smaller body of work asks whether such gains hold up rather than proposing new methods. Retrieval-augmented systems have been shown to be sensitive to linguistic variation in the question [29], and surveys of the area observe that the supporting evaluations rest on a narrow benchmark base [30, 31]. This stance has a long history in IR and NLP, where significance testing [33], bootstrap confidence intervals [34], and multiple-comparison correction [35, 36] separate real effects from noise. The present study belongs to this line and extends it in two respects: it isolates each method on a shared backbone, and it replaces the usual homogeneous-prose test with heterogeneous, mixed-format collections.

First-stage retrieval and reranking. The backbone shared across all conditions is the standard cascade of a dense bi-encoder [37, 38, 39] followed by a cross-encoder reranker [40, 41]; late-interaction retrievers such as ColBERTv2 [42] are an alternative we do not adopt, because we hold a single embedder fixed across systems. The strength of cross-encoder reranking is well established, but its interaction with the methods above is rarely measured, because those methods are usually evaluated on top of a weaker first stage or in isolation. Holding a strong reranker fixed while toggling each method is what lets the small remaining effects be attributed to the methods themselves (Section 7).

Benchmarks and the heterogeneity gap. QA and RAG benchmarks are predominantly homogeneous: open-domain and multi-hop questions over Wikipedia [15, 16, 17, 18, 43], with single-document scientific text [19, 20] and RAG-specific multi-hop sets [44] as the main variations. Aggregators such as BEIR [45] and KILT [46] broaden the tasks but keep each one single-modality, and recent work has begun to flag the absence of corpus-level, cross-document evaluation [32]. None of these mixes code, tables, PDFs, and prose within a single collection, and each ties relevance to a fixed passage segmentation. HetDocQA (Section 3) fills this gap, and its span-based labels make retrieval metrics independent of how each system chunks the corpus.

3 HetDocQA

HetDocQA is a question-answering benchmark over *heterogeneous, multi-format* document collections. Its purpose is to test whether RAG methods that help on homogeneous prose still help when a single collection mixes modalities and evidence is spread across documents.

Sources and modalities. The corpus comprises 258 documents grouped into 118 realistic mixed-format collections (for example, a paper PDF paired with its reference implementation and a documentation page). Documents span five modalities (PDF 92, prose 118, markdown 22, code 22, and tabular 4) and are drawn only from permissively licensed sources (MIT, Apache-2.0, BSD-3-Clause, CC-BY-SA-4.0, and arXiv e-prints) [47]. We release the benchmark as *pointers plus hash-verified build scripts* rather than relicensed content, so the corpus reconstructs deterministically from the original sources.

Chunker-agnostic span labels. Relevance is annotated as *character spans in the source documents*, not as chunks. At evaluation, each system chunks the corpus by its own segmentation; a retrieved chunk counts as relevant if it overlaps an evidence span by at least 50% (Figure 2). This makes retrieval metrics independent of chunking, so a system is never penalized or rewarded for a chunking choice that happens to align with the label granularity. Questions carry 2.18 evidence spans on average, and 527 of 762 require multiple evidence spans; this multi-span structure is what supports graded nDCG [48].

Table 1: HetDocQA composition. Questions by type and split (left), and the source corpus (right). Splits are disjoint by collection.

Type	Calib.	Dev	Test	Total
Factual	34	32	63	129
Code	37	34	64	135
Cross-document	52	48	91	191
Multi-hop	43	40	73	156
Thematic	38	41	72	151
All	204	195	363	**762**

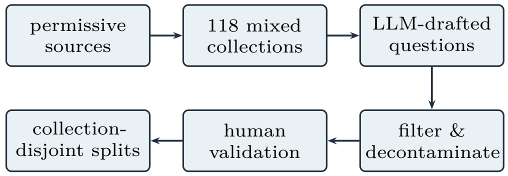
Corpus: 258 documents / 118 collections / 3,517 passages. Modalities: prose 118, PDF 92, markdown 22, code 22, tabular 4. Mean 2.18 evidence spans/question; 527 questions multi-evidence.

Construction and validation. Questions were drafted by a strong closed model that is not among the generators we evaluate, so that no evaluated system is scored on questions written in its own style. For the multi-hop and cross-document types, the model was given passages from different documents, so that answering the question requires combining them. Each draft then passed a sequence of automatic filters: questions answerable without any retrieved document were discarded, so that the benchmark measures retrieval rather than memorized knowledge; near-duplicate questions were removed by embedding similarity; and an automatic check confirmed that the cited evidence supports the answer and that the type label and phrasing are appropriate. A human reviewer then confirmed, for each released question, that it is answerable from its evidence, that the evidence spans are correct, and that the type label is right. This human pass is currently single-annotator; inter-annotator agreement has not yet been measured, and Section 8 lists a second annotation pass as planned work [49].

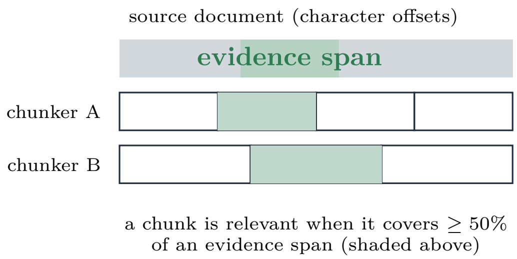
Splits and leakage control. The 762 questions are split *disjointly by collection* into calibration (204), dev (195), and test (363). Because splits share no collection, a threshold tuned on dev cannot exploit corpus structure that recurs in test. All tunable thresholds in every system are fixed on the calibration and development splits, and the test split is evaluated once. Table 1 summarizes the composition.

4 Methods and Hypotheses

This section describes the eight methods whose effect we measure and, for each, states the hypothesis under which it should help when the corpus is heterogeneous.



(a) Construction pipeline.



(b) Span→chunk relevance, computed per system at evaluation time.

Figure 2: HetDocQA construction and the fairness mechanism. (a) Permissive sources become mixed-format collections; a model distinct from any evaluated generator drafts questions, which are filtered, decontaminated, and human-validated, then split disjointly by collection. (b) Each evidence label is a character span in the source document. At evaluation, each system's own chunks count as relevant when they cover at least half of an evidence span, so retrieval metrics do not depend on chunking choices.

The methods are grouped by the stage of retrieval they act on: the query, the candidate pool, the ranking, and the answer decision. Most are established methods drawn from prior work and reproduced faithfully from their original descriptions, so that the comparison rests on the methods rather than on implementation differences. The one exception is the per-source corrector (SSCC), introduced here to address a difficulty specific to heterogeneous pools. Every method is evaluated on the shared backbone of Section 5, applied or withheld in isolation.

4.1 Query stage

Query expansion (HyDE, step-back). A dense bi-encoder matches a question to a passage by embedding similarity, which degrades when the question and the passage share little surface vocabulary. HyDE [4] addresses this by prompting the generator for a hypothetical answer passage and retrieving with the embedding of that passage rather than of the raw question; step-back prompting [5, 6] first abstracts the question to a more general form. Both are reproduced and applied as the default query transformation. *Hypothesis.* On heterogeneous data the vocabulary gap is wider, because the target evidence may be code or a table rather than prose, so a generated answer-like passage should land closer to the target in embedding space than the question does.

Modality-aware HyDE. A single prose hypothesis tends to retrieve prose. We extend HyDE to generate a hypothesis per target modality (for example a code snippet for a code question and a small table for a tabular question) and retrieve with each, so that the expansion can reach non-prose evidence. *Hypothesis.* Modality-matched hypotheses retrieve the modality that holds the answer, which a prose-only hypothesis misses. We report this method as an exploratory probe rather than a default.

4.2 Candidate-pool stage

Hierarchical summarization (RAPTOR). RAPTOR [7] recursively groups chunks and replaces each group with a generated summary, building a tree whose internal nodes are abstractions over many leaves. We follow the original design: soft clustering by a full-covariance Gaussian mixture over a UMAP projection [50] with the number of mixture components selected by BIC [51], recursion to at most three levels, and a collapsed-tree retrieval pool that contains leaves and summaries together. *Hypothesis.* Thematic and multi-hop questions ask for information that no single chunk states, so adding summary nodes that synthesize several chunks gives the retriever a candidate that matches such questions directly.

Cross-document tier. In the long-document question-answering settings for which RAPTOR was developed, the tree is constructed over the chunks of a single document, so its summary nodes aggregate evidence only within that document. We add a collection-level tier that applies the same soft clustering and summarization to the chunks of all documents in a collection, allowing a summary node to aggregate evidence drawn from several documents. This is closely related to the community summaries of GraphRAG [8], here adapted to RAPTOR-style soft clustering. To bound the cost of maintaining the tier under corpus updates, each node is keyed by a hash of its constituent chunk set, and only nodes whose membership changes are recomputed when a document is added. *Hypothesis.* Cross-document and thematic questions draw evidence from several documents simultaneously, so a summary node that already aggregates that evidence should be retrievable in cases where no single-document node matches.

Graph-augmented retrieval (GAHR). We construct a typed graph over chunks with four edge types: sequential edges between adjacent chunks, semantic edges between chunks whose cosine similarity exceeds

0.7, cross-reference edges derived from explicit references, and abstract-syntax-tree edges between structurally related code chunks. Node representations are refined by a GraphSAGE encoder [52] with type-specific message passing, trained without supervision using a neighbor-sampling objective; we additionally evaluate a deep graph infomax objective [53]. At query time, the top dense retrievals seed a personalized PageRank computation [54] over the typed graph, and the highest-mass nodes are added to the candidate pool within a fixed budget. The use of personalized PageRank for retrieval follows HippoRAG [9]; our construction is the particular combination of typed chunk-level edges, learned GraphSAGE refinement, and PageRank expansion, which is the variant evaluated here. *Hypothesis.* In heterogeneous collections, related evidence is often connected structurally rather than by surface similarity (a function and its documentation, or a table and the paragraph that discusses it), so graph expansion should recover such evidence even when it lies far from the query in embedding space.

4.3 Ranking stage

Cross-encoder reranker. The first-stage dense retriever returns a candidate pool that a cross-encoder then re-scores by jointly encoding the question with the candidates [40, 41]. We treat the reranker as a method and ablate it like the others, which lets us measure how much of the pipeline's quality it accounts for.

Dual-path retrieval with rank fusion (DPHF). Two retrieval paths, one from the raw question and one from the HyDE hypothesis, are combined by reciprocal rank fusion [14], which scores each candidate c by $\sum_p 1/(k + r_p(c))$ with k=60, where $r_p(c)$ is the rank of c in path p. A common dual-path variant merges the two paths by removing duplicates; here they are combined by rank fusion instead. *Hypothesis.* The two paths surface partly different relevant chunks, so fusing their rankings should recover more of the relevant evidence than either path alone.

Per-query routing (EGR and variants). Rather than always applying the same retrieval strategy, a router can choose a strategy per query. Our entropy-gated router (EGR) computes a softmax over the K nearest-neighbor distances, $p_i = \text{softmax}(-d_i/T)$, and routes on the entropy $H = -\sum_i p_i \log p_i$, which is bounded by $\log K$: low entropy to single-path semantic retrieval, intermediate entropy to fusion, and high entropy to step-back, with a multi-hop flag when entropy is high and the query mentions more than one entity. We compare EGR against a keyword router, a compositional router that combines several such signals, and an oracle that selects the best strategy per query using test labels, which upper-bounds any router. The router thresholds are fit on a held-out calibration split. *Hypothesis.* Different query types are best served by different strategies, and the local geometry of the query's neighborhood is a cheap signal for which strategy to use.

4.4 Answer-decision stage

Corrective retrieval (CRAG). CRAG [10] grades the retrieved evidence with an LLM judge and, when the evidence is judged weak, rewrites the query and retrieves again. It is reproduced here with a single confidence threshold on the judge score. *Hypothesis.* When the first retrieval is poor, detecting this and re-retrieving recovers the answer that a fixed pipeline would miss.

Per-source calibrated correction (SSCC, this work). The accept-or-reject decision in CRAG uses one threshold, but in our pipeline a candidate's score can come from different sources that live on different scales: the bi-encoder reports $1/(1 + L_2)$ similarities, whereas the cross-encoder reports reranker logits. A single threshold is therefore miscalibrated for at least one source. SSCC keeps a separate threshold per source, $\tau(s) = \tau_0\,(1 + \alpha\,\mathbf{1}[s = \text{bi-encoder}])$, with the thresholds fit per source from the score distributions of relevant and non-relevant passages on the calibration split. *Hypothesis.* Heterogeneous pools mix sources and modalities whose score scales differ, so calibrating the accept-or-reject decision per source should keep good evidence and drop bad evidence more accurately than one global threshold, and the benefit should grow with heterogeneity.

5 Experimental Setup

Shared backbone. Every system shares one bi-encoder (bge-m3 [39]), one reranker (jina-reranker-v3 [55]), and one generator (gpt-4.1-mini), all held fixed. jina-reranker-v3 is a strong listwise late-interaction reranker: it scores the candidate pool jointly in one context window rather than scoring each query–candidate pair independently. Because that joint scoring could in principle absorb part of what the pool-side methods aim to add, our conclusions about those methods are, strictly, conditional on this reranker; a preliminary check with a standard pairwise cross-encoder (bge-reranker-v2-m3), which scores each query–candidate pair independently, confirms it reorders the candidate pool substantially and is itself a strong reranker, but a full per-method ablation under it is left to future work (Section 8). The generator is different from the model that drafted HetDocQA and from the faithfulness judge, so that no evaluated system is favored by the model that wrote the questions or by the one that judges faithfulness. Retrieval runs over a LanceDB index, and all embeddings, reranks, and generations are cached, so the full ablation suite replays offline and deterministically.

Methods and ablation protocol. The eight methods are those of Section 4: query expansion (HyDE, step-back) and its modality-aware variant; hierarchical summarization (RAPTOR) and the cross-document tier; graph-augmented retrieval (GAHR); the cross-encoder reranker; dual-path rank fusion (DPHF); per-query routing (EGR, keyword, compositional, and an oracle upper bound); corrective retrieval (CRAG); and per-source calibrated correction (SSCC). Each method is applied or withheld independently, so every ablated run differs from the full pipeline in exactly one method. We report the full pipeline and, for each method, the pipeline with that method removed.

Metrics. We report nDCG@10 (graded) [48] and Success@10 for retrieval, and exact match and token-F1 for answers. Headline comparisons use token-F1. Faithfulness, where reported, is atomic-claim support judged by a model distinct from all tested generators [56, 57, 58].

Statistics. Every reported metric carries a 95% bootstrap confidence interval ($\geq$10k resamples) [34]. Each method is compared to the full pipeline with a paired bootstrap test [33]. Across each benchmark's family of method ablations we apply Holm–Bonferroni correction [35] (BH–FDR [36] agrees on every conclusion). We call an effect "reliable" only when its corrected comparison rejects at α=0.05 *and* its sign is consistent across dev and test.

6 Results

6.1 Effect of each method on HetDocQA

Table 2 reports the ablation on the HetDocQA test split. Three observations follow from it.

The reranker accounts for most of the pipeline's quality. Removing it collapses retrieval almost entirely (nDCG@10 falls from 0.644 to 0.034 and Success@10 from 0.87 to 0.08) and answer F1 drops by 30 points, an effect no other method approaches within an order of magnitude. This nDCG is lower than semantic-only retrieval (0.288) because the ablated pipeline still expands the candidate pool with HyDE, hierarchy, and graph nodes but, without the cross-encoder, orders that enlarged multi-source pool by raw, cross-incomparable scores; the collapse therefore measures the loss of ordering over an inflated pool rather than implying that the reranker alone supplies retrieval quality. Figure 3 shows the same collapse as a Recall@k curve, where the

Table 2: Headline ablation, HetDocQA test (n=363). Each row removes one method from the full pipeline. ΔF1 is relative to FULL; p is a paired bootstrap test on F1. ✓ significant after Holm–Bonferroni *and* keeps its sign on dev; ○ significant on test but reverses sign on dev; ✗ no reliable effect. nDCG and F1 are absolute.

Configuration	nDCG@10	F1	p	
FULL	0.644	0.548	n/a	
−reranker	0.034	0.249	< .001	✓
−HyDE	0.542	0.481	< .001	✓
−SSCC	0.602	0.523	.005	✓
−RAPTOR	0.650	0.523	.004	○
−graph (GAHR)	0.641	0.542	.214	✗
−cross-document	0.644	0.545	.213	✗
−CRAG	0.644	0.546	.478	✗
−DPHF (RRF)	0.644	0.548	1.00	✗
semantic only	0.288	0.421	< .001	

cross-encoder, rather than pool expansion, is what lifts recall.

Among the eight methods, only query expansion (HyDE) and the per-source corrector (SSCC) produce reliable improvements. Removing HyDE costs 6.7 F1 points (p<10^{-3}) and removing SSCC costs 2.5 points (p=0.005); both remain significant after Holm correction and retain their sign on the dev split.

Every reranking or pool-expansion method (rank fusion, graph expansion, the cross-document tier, and corrective re-retrieval) changes F1 by less than half a point, and none is significant after correction. The single ambiguous case is RAPTOR. Removing it is significant on test (p=0.004), but the sign reverses on dev, where removing it improves F1 (from 0.493 to 0.507, p=0.114). Under our consistency rule this does not count as a reliable gain; we treat it as unstable and return to it in Section 7.

6.2 Replication across benchmarks

The same ablation on two standard benchmarks (Table 3) shows the same pattern. On MuSiQue [17] (multi-hop Wikipedia), only the reranker and HyDE are significant after correction. On QASPER [19] (single-document scientific prose), only the reranker is significant. On neither does any pool-expansion or reranking method, nor SSCC, produce a reliable gain.

6.3 Significance across benchmarks

Table 3 collects the outcomes across the three benchmarks and eight methods. The cross-encoder reranker is the only one that helps on all three. Query expansion helps where the query and the corpus share little surface vocabulary (MuSiQue and HetDocQA) but not on QASPER. SSCC, introduced here, helps only on Het-

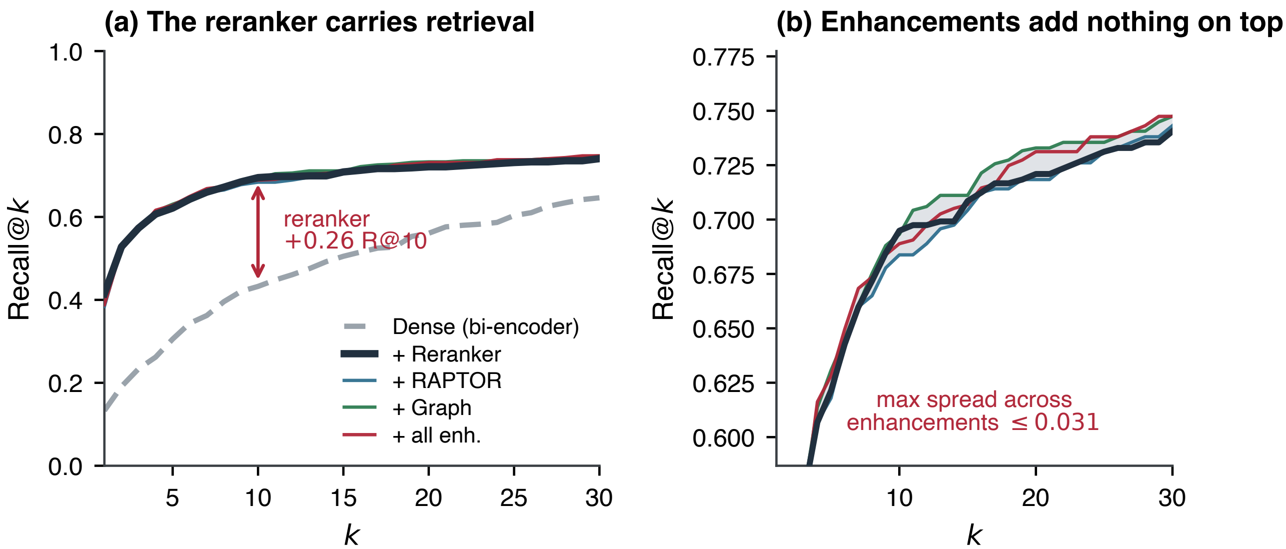


Figure 3: Effect of the reranker on recall. Recall@k on HetDocQA dev, with all configurations sharing the same first-stage retriever. Adding the cross-encoder reranker raises the curve sharply, whereas adding the pool-expansion methods (RAPTOR, graph, and all combined) on top of the reranker changes it little.

Table 3: Cross-benchmark significance. ✓: improvement is significant after Holm–Bonferroni with consistent sign; ✗: no reliable effect; ○: significant but sign-unstable across splits. The reranker helps on all three benchmarks, while the only novel method that reaches significance (SSCC) helps on HetDocQA alone.

Method	MuSiQue	QASPER	HetDocQA
Cross-encoder reranker	✓	✓	✓
HyDE (query expansion)	✓	✗	✓
SSCC (per-source corr.)	✗	✗	✓
RAPTOR hierarchy	✗	✗	○
DPHF / RRF fusion	✗	✗	✗
GAHR graph expansion	✗	✗	✗
Cross-document tier	✗	✗	✗
CRAG correction	✗	✗	✗

DocQA. No other method is significant after correction on any benchmark.

6.4 Per-query routing

A natural objection is that the methods should be applied *selectively*, per query. We test this directly with a routing triad: keyword routing, an entropy-gated router (EGR), a compositional router, and an *oracle* that picks the best strategy per query with test labels. On all three benchmarks, no realizable router beats always choosing the single best fixed strategy in answer F1, and the entropy-gated router is *worse* than the trivial baseline (it collapses onto step-back). Even the oracle's F1 gain is small, and on HetDocQA the oracle improves nDCG without improving F1, which suggests the better-ranked evidence does not, with this generator, change the generated answer. Figure 4 shows the cause: the signal EGR routes on, the entropy of the nearest-neighbor distance distribution, is nearly constant across queries and sits close to its $\log K$ ceiling, so there is almost nothing to route on.

6.5 SSCC is heterogeneity-specific

SSCC keeps a separate confidence threshold per score source, because the bi-encoder and cross-encoder relevance scores live on different scales (Figure 6). A single global threshold mis-calibrates one of them. On HetDocQA the distinction matters: SSCC is significant on both dev (p=0.011) and test (p=0.005), and remains so after correction. On the homogeneous controls it has no significant effect (MuSiQue p=0.35, QASPER p=0.08). The benefit appears only when the candidate pool mixes modalities whose scores lie on different scales, and grows with the heterogeneity of the benchmark (Figure 5). A lighter alternative we do not evaluate is to map the two score sources onto a common scale (for example by per-source normalization) before applying one global threshold; whether per-source thresholding retains an advantage over such normalization is left to future work.

7 Discussion

Why most methods do not help. The methods that show no reliable effect have a common explanation. A strong cross-encoder reranker, applied to a candidate pool that contains the relevant evidence, already places that evidence near the top of the ranking. Three consequences follow, each consistent with the measurements.

First, reranking the same pool has little room to

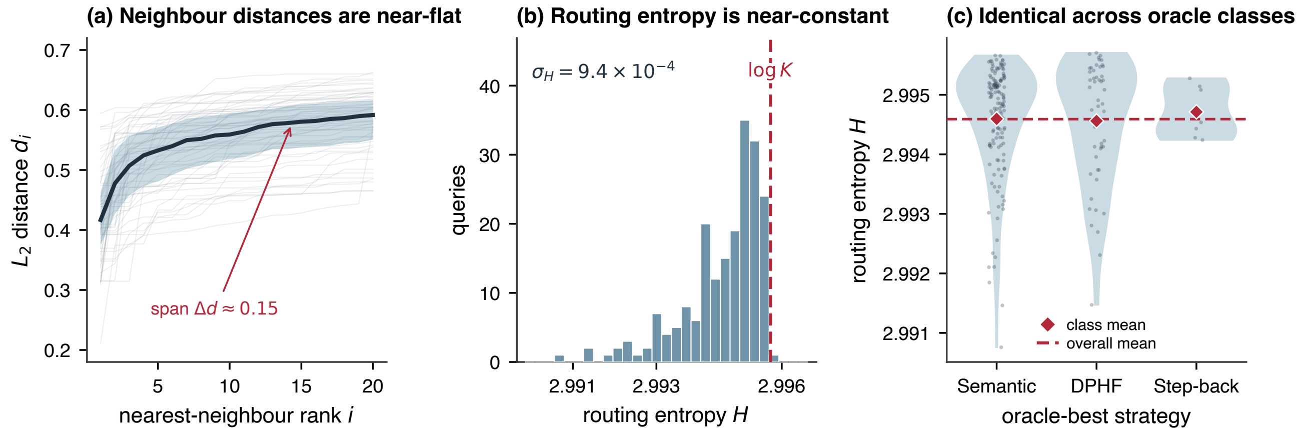


Figure 4: Routing degeneracy. Per-query nearest-neighbor distances are nearly identical across queries (left); the entropy that EGR routes on sits at its theoretical ceiling with very little variance (center); and the per-query best strategy gives almost no advantage over a single fixed strategy (right). The signal a per-query router would need is not present.

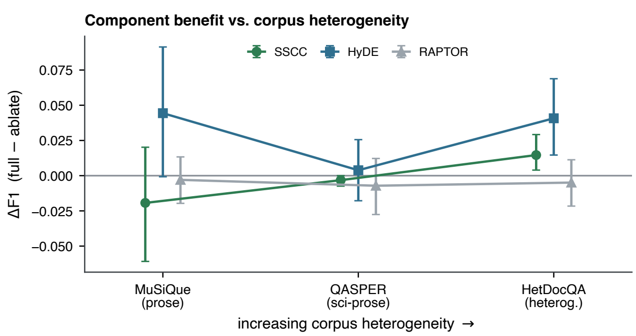


Figure 5: The gain is heterogeneity-specific. Effect size (paired ΔF1 with 95% CI) of SSCC, HyDE, and RAPTOR as the benchmark moves from homogeneous multi-hop prose (MuSiQue) through single-document scientific prose (QASPER) to the heterogeneous collection (HetDocQA). SSCC's interval excludes zero only on HetDocQA.

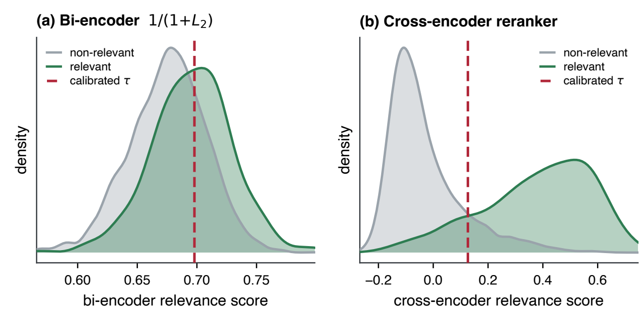


Figure 6: Why one threshold fails. Relevance-score distributions for relevant vs. non-relevant passages under the bi-encoder (left) and the cross-encoder (right) on HetDocQA. The two sources separate relevant from non-relevant at very different score ranges, so the per-source thresholds SSCC fits (dashed) differ substantially; a single global threshold cannot be right for both.

help. Per-query routing, rank fusion, graph rescoring, and corrective re-grading all reorder or re-weight candidates that the reranker has already ordered well, so they cannot move the top of the list much, and none is significant after correction (Table 3). Second, expanding the pool with abstractions helps little. The summary nodes added by RAPTOR and by the cross-document tier compete against the specific chunk that contains the answer, which the reranker tends to prefer; when a summary is slightly off topic it can even displace that chunk, which is a plausible source of the RAPTOR sign change between dev and test. Third, the gains that remain are at the input and at the answer decision, because those are the only two places that change something other than the ordering of an already well-ordered pool: query expansion changes what enters the pool, and SSCC changes which candidates are accepted after ranking. These are the two methods with a significant effect.

This explanation is conditional on the reranker. A weaker first stage or a weaker reranker could leave room for the pool-side methods to help; that regime is left to future work.

A query-side probe. If the gains that remain lie at the input, then query expansion aware of the target modality should help on heterogeneous data. We tested a modality-aware HyDE variant (Section 4). It gives the best point estimates on code and tabular questions, but at n=56 per slice the difference from a plain multi-prose HyDE ensemble is not significant (Figure 8). We therefore report it as a promising but inconclusive direction rather than a result.

Validity of the null results. A null result is informative only if the method was genuinely active during evaluation. For each reimplemented method we therefore verified that it alters the retrieved candidate pool as designed and that it operates within the latency budget of the full experiment suite. Graph expansion via personalized PageRank, for instance, contributes graph-reachable nodes to the pool and runs in about 0.15 s per query using a cached sparse transition operator. With these checks in place, the null effects in Table 3 are not an artifact of inactive implementations, though we cannot exclude that more aggressive per-method tuning would change them.

8 Limitations

Several boundaries scope these results. HetDocQA contains 762 questions, sized for clean human validation rather than web scale, so the smallest per-type slices (such as the n=56 modality probe) have limited statistical power. More broadly, because several of our central claims are null results, their reach is bounded by statistical power: at n=363 on the test split the paired comparisons resolve the multi-point effects of the reranker and query expansion cleanly, but not the sub-point differences the pool-expansion methods produce, so those should be read as "no gain detectable at this sample size" rather than as proof of exactly zero effect; a tighter bound would need a larger test set. The findings are conditional on a strong reranker, and specifically on a listwise one whose joint scoring of the candidate pool could itself absorb part of what the pool-side methods aim to add; a preliminary check confirms a pairwise cross-encoder is also a strong, differently-behaved reranker, but replicating the full per-method ablation under it, and under a weaker first stage, is left to future work. A single generator is used in the controlled rows, and a substantially weaker or stronger one might change which retrieval gains reach the answer. Validation of HetDocQA is currently single-pass, without inter-annotator agreement, and a second annotation pass is planned before archival release. These boundaries limit how far the results generalize; they do not affect the comparisons themselves, which hold the backbone fixed across conditions. The benchmark and harness are released so that the comparisons can be repeated and contested.

9 Conclusion

HetDocQA makes it possible to ask which retrieval methods help when a collection mixes code, markdown, tables, prose, and PDFs. Across HetDocQA, MuSiQue, and QASPER, two methods yield reliable gains: query expansion, and SSCC, a per-source calibrated corrector that addresses the incomparable score scales a heterogeneous pool produces. The reranking and pool-expansion methods in wide use give no reliable gain once a strong cross-encoder reranker is present. HetDocQA, the evaluation harness, and the reproduction scripts are released so that further methods can be tested on heterogeneous data under the same controls.

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A Appendix: additional analyses

This appendix reports two analyses that support the main text but are not central to its argument: a characterization of where HetDocQA is difficult, and the full three-arm comparison behind the query-side probe of Section 7.

A.1 Difficulty by question type and modality

Figure 7 characterizes where HetDocQA is hard, using the full pipeline and independent of any single method. Answer quality varies sharply by question type (panel a): factual questions are answered most accurately, whereas thematic questions, which ask for a synthesized position rather than a single fact, are the hardest, with multi-hop and cross-document questions in between. Retrieval quality varies by the modality of the evidence (panel b): evidence in code and markdown is located most reliably, and evidence in prose the least, despite prose being the modality on which the underlying methods were originally developed. These two axes are the backdrop for the per-method results of Section 6: the benchmark is not uniformly difficult, so a method that helped only on its easy region would not produce a reliable aggregate effect.

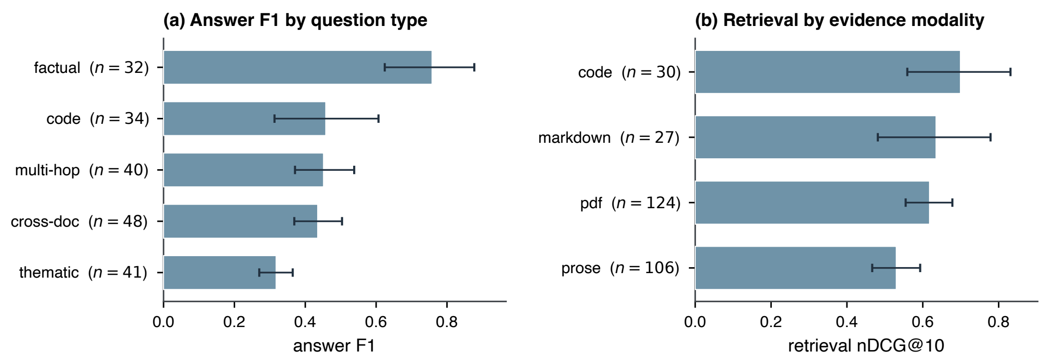

Figure 7: Difficulty by question type and modality. Answer F1 by question type (panel a) and retrieval nDCG@10 by the modality of the evidence (panel b) on HetDocQA, under the full pipeline. Bars show the mean with a 95% confidence interval; n is the number of test questions in each group.

A.2 Modality-aware query expansion

Figure 8 reports the three-arm comparison from Section 7, restricted to the 56 code and tabular questions on which a modality-aware hypothesis could plausibly help. Three forms of query expansion are compared: a single generic prose hypothesis, an ensemble of three prose hypotheses, and an ensemble of three modality-matched hypotheses (for example a code snippet for a code question). The modality-matched variant has the best point estimate on every metric, but its margin over the prose ensemble is within noise at this sample size. Separating the benefit of matching the modality from the benefit of simply drawing more hypotheses would require a larger code- and table-heavy evaluation set; we therefore report the direction as promising rather than established.

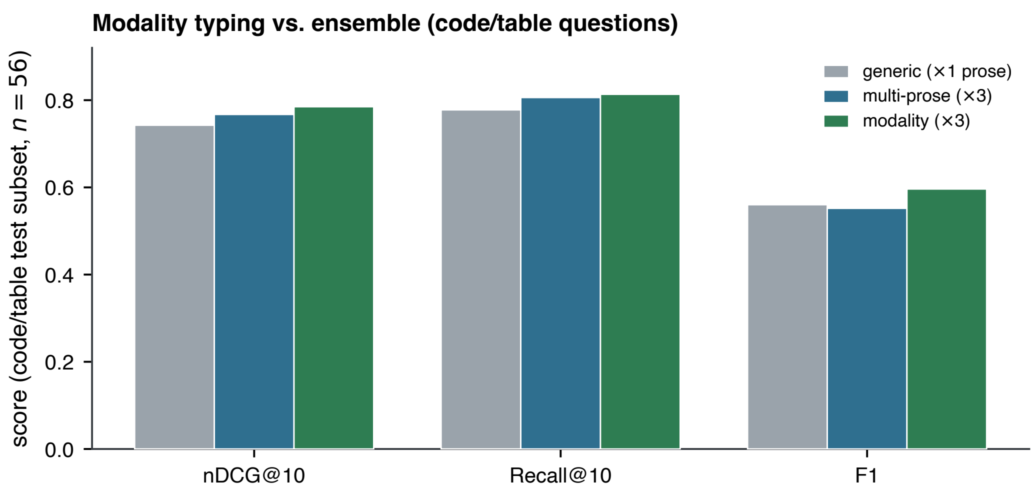

Figure 8: Modality-aware query expansion, three-arm comparison. On the 56 code and tabular questions, the modality-matched ensemble gives the best point estimate on nDCG@10, Recall@10, and answer F1, but its difference from the prose ensemble is not significant at this sample size.